

AP Statistics: The Complete Exam Guide

The 5 units explained, the four statistical practices, a free-response playbook for inference, and 10 hard practice questions with full explanations.

Hi, I'm **Pigo!** I'll drop tips all through this guide.

**3 hrs**

90 min multiple choice
+ 90 min free response

42+4

multiple-choice +
free-response questions

Calc

graphing calculator
with stats functions

60–62%

scored 3 or higher
in recent years

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01 THE EXAM AT A GLANCE

Two sections, one score

Your AP score (1–5) combines a multiple-choice section and a free-response section, each worth half. A graphing calculator is allowed throughout, and Bluebook has a built-in Desmos calculator.

50%

Section I • Multiple choice • 42 questions

50%

Section II • Free response • 4 questions

Exam structure

Section	Weight & timing	What you do
I • Multiple choice	50% of score 1 hr 30 min 42 questions	Questions across all 5 units, testing the 4 statistical practices. Many use data displays, study descriptions or computer output. Calculator permitted.
II • Free response	50% of score 1 hr 30 min 4 questions	Q1 formulating questions and collecting data • Q2 analyzing data and interpreting results • Q3 inference-focused • Q4 multi-content question that combines ideas. Calculator permitted.

FULLY DIGITAL

Everything happens in Bluebook

You answer every question, including the written responses, in College Board's **Bluebook** testing app. Responses are submitted automatically at the end. Practice typing timed answers before May.

REFERENCE MATERIALS

Formulas and tables are provided

A formula sheet and statistical tables (normal, t, chi-square) are available. Knowing **when** to use each formula matters more than memorizing it.

How to use this guide

Just starting?

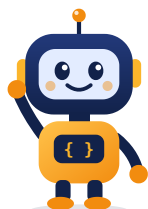
Read pages 2–6, then follow the roadmap on page 8 month by month.

Mid-year?

Check the traps on page 7 and drill the free-response playbook on page 6.

Exam in weeks?

Take the 10 questions on page 9 in 22 minutes, then use the final-four-weeks plan.



Pigo's tip: Statistics is a writing course in disguise. Free-response answers are scored on **communication**: context, conditions and clear conclusions.

SECTION I

02 The multiple-choice section

42 questions in 90 minutes gives you a little over 2 minutes per question. Many questions are about interpreting studies and output, not heavy calculation.

42

questions with four answer choices

~2:10

minutes per question on average

50%

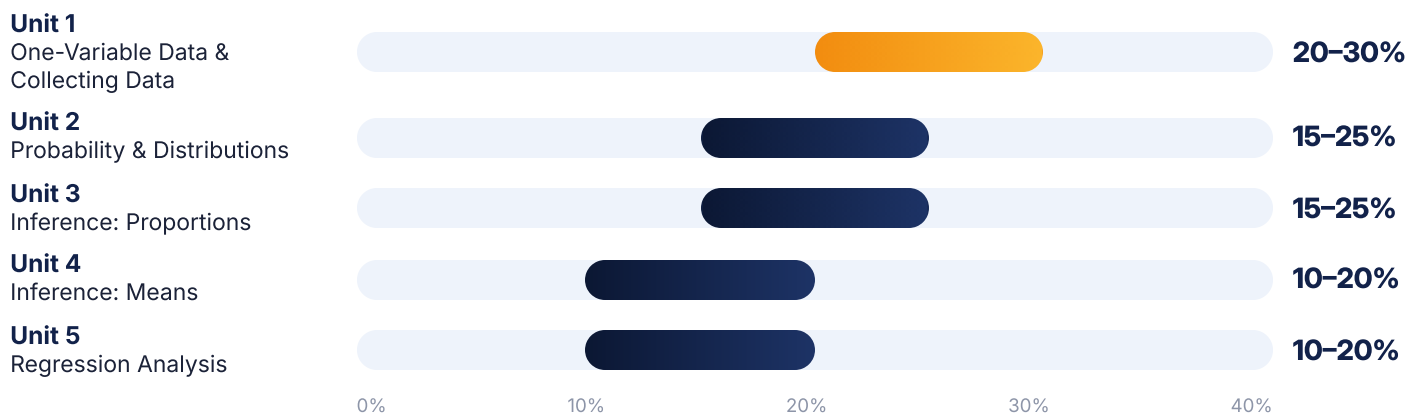
of your total AP score

0

points lost for a wrong answer, so never leave one blank

How the questions are weighted by unit

College Board gives a range for each unit's share of the exam. Bars show minimum to maximum.

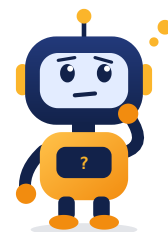


What this means for your study time: Unit 1, describing data and designing studies, is the biggest unit and underpins everything else. Inference (Units 3–4) together is a similar share.

Strategies that work on this section

- Identify the procedure first: one or two samples? proportions or means? test or interval?
- Watch the wording of interpretations. "95% confident" describes the method, not a 95% chance for this interval.
- Sketch the normal curve and shade before computing any probability.
- Read computer output carefully: slope, intercept, s , r and r^2 , and what each means in context.
- Eliminate absolute words like "proves" and "always" in study-design questions.
- Flag and move on after 3 minutes. Return once the quick points are banked.

Correlation is **not** causation. Only a randomized experiment supports a cause-and-effect conclusion.



03 WHAT THE COURSE COVERS

The 5 units, in plain English

Everything on the exam fits into these five units.

UNIT

1

20–30%

Exploring One-Variable Data & Collecting Data

Describing distributions and designing good studies.

Be ready to: describe shape, center, spread and unusual features in context; use z-scores and the normal model; identify sampling methods and bias; design experiments with random assignment, control and blocking; know what conclusions each design supports.

UNIT

2

15–25%

Probability, Random Variables & Distributions

Chance, simulation and the distributions that model it.

Be ready to: use probability rules and two-way tables; test independence; run simulations; find the mean and SD of discrete random variables; use binomial models; describe sampling distributions.

UNIT

3

15–25%

Inference for Categorical Data: Proportions

Intervals and tests for proportions, plus chi-square.

Be ready to: check conditions; build and interpret confidence intervals; carry out significance tests for one and two proportions; interpret p-values, Type I and II errors and power.

UNIT

4

10–20%

Inference for Quantitative Data: Means

Intervals and tests for means using t-procedures.

Be ready to: choose between one-sample, paired and two-sample t-procedures; check conditions; build intervals and run tests; interpret results in context.

UNIT

5

10–20%

Regression Analysis

Relationships between two quantitative variables.

Be ready to: describe scatterplots; interpret slope, intercept, r and r^2 ; use residuals and residual plots; identify influential points; make predictions and avoid extrapolation.

◆ The four statistical practices

Formulate questions

Pose investigative questions that data can answer.

Collect data

Plan sampling and experiments that avoid bias.

Analyze data

Describe patterns, model chance and compute statistics.

Interpret results

Justify conclusions in context with evidence.

SECTION II

04

The 4 free-response questions

90 minutes, 4 questions. Each question is worth 10 points, with credit for each part, so complete, clear communication matters.

Q1

Questions & collecting data

Multi-part

Pose an investigative question, then plan a sample or experiment and explain how it avoids bias.

Name the treatments, random assignment and response variable explicitly.

Q2

Analyzing & interpreting

Multi-part

Describe distributions or relationships, compute probabilities or statistics, and interpret them in context.

Compare center, spread, shape and outliers using comparative words.

Q3

Inference

Full procedure

Carry out a test or interval: state hypotheses and parameter, name the procedure, check conditions, calculate, and conclude in context.

Link the p-value to α and to the claim in one sentence.

Q4

Multi-content

Combines ideas

Pulls together several units in an unfamiliar setting. Rewards careful reasoning more than recall.

Read the whole question before starting; later parts often hint at earlier ones.

• A time plan for the 90 minutes

Q1

~20 min

Q2

~22 min

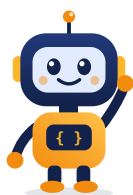
Q3

~24 min

Q4

~24 min

A suggested split, not a rule. Start with the question type you're strongest at, and never leave a question blank.



Every answer needs **context**. "The mean is 12" scores less than "the mean wait time is 12 minutes".

05

FREE-RESPONSE PLAYBOOK

Anatomy of a full-credit answer

Inference questions follow a four-step structure. Here is a short practice prompt with a full-credit answer.

PRACTICE TASK • WRITTEN BY ACT SAT GO

Practice prompt (Q3 style): A school claims that 30% of students walk to school. In a random sample of 200 students, 72 walk. Do the data give convincing evidence at $\alpha = 0.05$ that the proportion who walk is greater than 0.30?

Full-credit response

- ① $H_0: p = 0.30$, $H_a: p > 0.30$, where p = proportion of all students at the school who walk.
- ② One-sample z-test for a proportion. Random sample ✓; $200(0.3) = 60 \geq 10$ and $200(0.7) = 140 \geq 10$ ✓; $200 < 10\%$ of all students at the school (assume 2000+) ✓.
- ③ $\hat{p} = 0.36$, $z = (0.36 - 0.30)/\sqrt{(0.3 \cdot 0.7/200)} \approx 1.85$, p-value ≈ 0.032 .
- ④ Because $0.032 < 0.05$, we reject H_0 . There is convincing evidence that **more than 30% of students at this school walk**.

- ① **Hypotheses + parameter** defined in words.
- ② **Name the test and check every condition** with numbers.
- ③ **Show the test statistic and p-value** with the formula.
- ④ **Conclude in context**, linking p-value to α .

Five habits for every free-response question

1

Define every parameter and variable in context.

2

Check conditions with numbers, not just names.

3

Show formulas, not only calculator results.

4

Never say a test "proves" or "accepts" H_0 .

5

Stuck? Name the procedure; that alone can earn credit.

06 LEARN FROM OTHERS

10 traps that cost points

Meet **Glitch**. He lives in these ten mistakes. Squash each one before May.

I **accept** the null!
(Don't.)



01 No context

Every description and conclusion must mention the variable and the population in the problem.

02 Wrong CI interpretation

"We are 95% confident the interval captures the true proportion," not "95% of students..." or "95% chance".

03 "Accepting" the null

You fail to reject H_0 . A large p-value is not evidence that H_0 is true.

04 Causation from observation

Only randomized experiments support cause-and-effect conclusions.

05 Conditions without numbers

"Large counts" needs $n\hat{p} \geq 10$ and $n(1 - \hat{p}) \geq 10$ shown with values.

06 Paired vs two-sample

Before-and-after on the same subjects is paired. Use the differences.

07 Practical significance

A tiny p-value means a result is unlikely to be chance alone, not that the effect is large or important.

08 Residual sign

Residual = actual – predicted. A negative residual means the model overestimated.

09 Bias vs variability

A bigger sample reduces variability, not bias from a flawed sampling method.

10 Calculator-speak

"normalcdf(...)" alone may not earn credit. Name the distribution, parameters and boundary.

Quick self-check: a regression on 30 cities gives $r^2 = 0.81$ and a negative slope. What is r , and what does r^2 mean in context? ($r = -0.9$; 81% of the variation in y is explained by the linear model with x .)

07

PLAN YOUR YEAR

Your study roadmap to May



A steady plan beats a last-minute rush. Shift the months to match your teacher's pacing, but keep practicing written explanations throughout.

Sep – Oct

Unit 1



Describing data & study design

biggest unit

Distributions, normal model, sampling and experiments. Write one "describe and compare" paragraph each week.

Nov – Dec

Unit 2



Probability & random variables

Probability rules, simulation, random variables, binomial models, sampling distributions.

Jan – Mar

Units 3–4



Inference for proportions and means

Intervals and tests, Type I/II errors, power and chi-square. Do one full inference question every week.

April

Unit 5 + practice



Regression and timed sections

Regression and residual analysis, then full timed multiple-choice and free-response sets.

Early May

Sharpen



Communication polish

Rewrite weak answers using rubric language. Review the traps list.

Exam day

May



AP Statistics exam

Sleep well, charge your device, pack your calculator, and use the checklist at the end.

◆ The final four weeks

Week	Focus	Target
4 weeks out	Diagnose: 21 multiple-choice questions plus 2 free-response questions	List your two weakest units
3 weeks out	Drill inference: one test and one interval per day	All four steps every time
2 weeks out	Full 3-hour practice exam	Finish both sections on time
Final week	Light review: interpretations, conditions, traps list	Rest the night before

08 TEST YOURSELF 10 practice questions

Original questions written by ACT SAT GO in the style of the exam, pitched at the harder end. Give yourself **22 minutes** for all ten, then check page 11.

1 UNIT 1 • NORMAL MODEL

Test scores are approximately normal with mean 70 and standard deviation 8. About what percent of scores are above 82?

- A 1.5%
- B 6.7%
- C 13.4%
- D 43.3%

2 UNIT 2 • SAMPLING DISTRIBUTIONS

In a population, 30% of adults prefer brand A. For random samples of 200 adults, about what is $P(\hat{p} > 0.35)$?

- A 0.006
- B 0.061
- C 0.122
- D 0.35

3 UNIT 2 • BINOMIAL

Each of 5 independent free throws succeeds with probability 0.2. What is the probability of **at least one** success?

- A 0.200
- B 0.328
- C 0.410
- D 0.672

4 UNIT 2 • TWO-WAY TABLES

Students were asked which of two lunch options they prefer.

	Option A	Option B	Total
Grade 9	30	20	50
Grade 10	45	55	100

Which statement is true?

- A $P(\text{Grade 10} \mid A) = 0.45$, and grade and preference are independent
- B $P(A \mid \text{Grade 10}) = 0.60$, and they are independent
- C $P(\text{Grade 10} \mid A) = 0.60$, and grade and preference are not independent
- D $P(A) = 0.60$, and they are not independent

5 UNIT 3 • CONFIDENCE INTERVALS

A 95% confidence interval for a population proportion is (0.42, 0.50). Which statement is correct?

- A If we repeated the sampling many times, about 95% of intervals made this way would capture p
- B There is a 95% probability that p is between 0.42 and 0.50
- C 95% of sample proportions fall between 0.42 and 0.50
- D p is definitely between 0.42 and 0.50

6 UNIT 3 • ERRORS & POWER

A researcher increases the sample size of a test while keeping $\alpha = 0.05$. Assuming H_0 is false, what happens?

- A $P(\text{Type I error})$ increases
- B Power decreases
- C $P(\text{Type II error})$ decreases and power increases
- D The p -value is guaranteed to fall below 0.05

08 TEST YOURSELF (continued)

7 UNIT 5 • RESIDUALS

The least-squares line is $\hat{y} = 12 + 3.5x$. For $x = 8$, the observed value is 38. What is the residual, and what does it mean?

- A -2; the line overestimates
- B 2; the line underestimates
- C -2; the line underestimates
- D 40; the predicted value

8 UNIT 5 • CORRELATION

A scatterplot shows a negative linear association with $r^2 = 0.81$. What is the correlation r ?

- A 0.81
- B -0.81
- C 0.90
- D -0.90

9 UNIT 1 • EXPERIMENTAL DESIGN

Researchers test a new fertilizer on 40 plots. Half the plots are on a hillside and half in a valley. Which design best controls for location?

- A Randomly assign fertilizer to 20 plots overall
- B Give fertilizer to all hillside plots
- C Block by location, then randomly assign treatments within each block
- D Let farmers choose which plots get fertilizer

10 UNIT 4 • CHOOSING A PROCEDURE

Each of 15 students takes a pretest and a posttest. Which procedure estimates the mean improvement?

- A Two-sample t-interval for $\mu_1 - \mu_2$
- B Two-proportion z-interval
- C Chi-square test for independence
- D Paired t-interval for the mean difference

09

CHECK YOUR WORK

Answer key & explanations

Read every explanation, even for questions you got right. Each one points to a pattern that keeps coming back on the exam.

#	Answer	Why
1	B • 6.7%	$z = (82 - 70)/8 = 1.5$. $P(Z > 1.5) \approx 0.067$. Choice C doubles it as if it were two-sided.
2	B • 0.061	SD of $\hat{p} = \sqrt{(0.3 \cdot 0.7)/200} \approx 0.0324$. $z = 0.05/0.0324 \approx 1.54$, so $P \approx 0.061$. Large counts: 60 and 140 ≥ 10 .
3	D • 0.672	$P(\text{at least one}) = 1 - P(\text{none}) = 1 - 0.8^5 = 1 - 0.328 = 0.672$.
4	C	$P(\text{Grade 10} \mid A) = 45/75 = 0.60$. $P(A) = 0.50$ but $P(A \mid \text{Grade 9}) = 0.60$, so the variables are not independent.
5	A	Confidence describes the method's long-run capture rate. Choices B and D treat this one interval as random or certain.
6	C	A larger sample reduces variability of the statistic, so a false H_0 is rejected more often: Type II error falls and power rises. α stays at 0.05.
7	A • -2	Predicted $\hat{y} = 12 + 3.5(8) = 40$. Residual = $38 - 40 = -2$. Negative residual: the line overestimated.
8	D • -0.90	$r = \pm\sqrt{0.81} = \pm 0.9$, with the sign of the slope. The association is negative, so $r = -0.9$.
9	C	Blocking groups similar plots so location differences don't hide the treatment effect; randomizing within blocks keeps the comparison fair.
10	D	Both scores come from the same students, so the data are paired: use a t-interval on the differences. Choice A wrongly treats the samples as independent.

9–10

Strong. Move on to full timed free-response practice.

6–8

On track. Revisit the unit behind every miss on page 4.

0–5

Build foundations now. Re-learn describing distributions and the four inference steps.

10 KNOW THE TARGET How students score

In recent years, about 60–62% of students earned a 3 or higher on AP Statistics, and about 17% earned a 5. Steady preparation is what moves a score up a full point.

60–62%

of students earned a 3 or higher in recent years

17%

earned the top score of 5 in recent years

3+

is the score many colleges accept for credit or placement

What each AP score means

5

Extremely well qualified

4

Very well qualified

3

Qualified

2

Possibly qualified

1

No recommendation

Ranges reflect recent College Board score distributions and shift a little each year. Credit and placement policies vary by college, so check each college's AP policy.

Exam-day checklist

- ☐ **Bluebook** is installed and updated, and your device passed the exam check
- ☐ Device **fully charged**, with the charger packed
- ☐ Admission ticket and photo ID, as your school requires
- ☐ Confirm your **exam day and start time** in May with your AP coordinator
- ☐ An **approved graphing calculator** with statistical functions and fresh batteries
- ☐ Plan your time: about **2 min** per multiple-choice question and **22 min** per free-response question
- ☐ Answer **every** multiple-choice question. Wrong answers don't lose points
- ☐ Use the provided **formula sheet and tables**

In the free-response section: answer in context, check conditions with numbers, show the formula behind each calculation, and end inference with a conclusion about the population.



ACT SAT GO • 1-ON-1 AP TUTORING

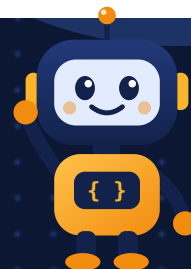
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Book a free diagnostic lesson. We'll find your weakest units, review your practice responses, and give you a week-by-week plan to exam day in May.

Book a free diagnostic lesson →



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Concept tutoring

Live 1-on-1 sessions on all five units, focused on choosing the right procedure.

Free-response coaching

Practice inference and multi-part questions, marked for communication like the real rubric.

Timed exam practice

Full-length multiple-choice sets with score reports, so you see progress before May.

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Web www.actsatgo.com

Email info@actsatgo.com

Phone +1 (945) 391-6179

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